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Semester – V

Business Analytics

Experiment No.: 2

Title: SUPERSTORE SALES ANALYTICS USING POWER BI

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SUPERSTORE SALES ANALYTICS USING POWER BI

GITHUB Link: <https://github.com/pragathidevanga/Business-Analytics>

Problem Statement

You are a Data Analyst asked to build a retail sales reporting solution for a US-based Superstore chain. The raw transactional data is available as a single flat file (CSV) containing order, customer, product, and shipping details for individual line items. The task is to import this flat file into Power BI, clean and transform the data using Power Query, build a suitable data model, and design an interactive dashboard for business reporting on sales, profit, and quantity across categories, regions, and time.

Methodology / Steps Followed

Step 1: Data Collection

- Source file: Sample_ Superstore.csv, a single flat-file dataset (comma-delimited, UTF-8 encoding) containing US retail order-line data.
- The raw file contains 19 columns: Row ID, Order ID, Order Date, Ship Date, Ship Mode, Customer ID, Segment, Country, City, State, Region, Product ID, Category, Sub-Category, Product Name, Sales, Quantity, Discount, Profit.
- 9,994 order-line records loaded, spanning Order Dates from January 2014 to December 2017.
- Unlike a multi-table database scenario (separate Products / Stores / Sales / Customer tables), this dataset arrives fully denormalized — product, customer, order, and location attributes are already combined into one flat table.

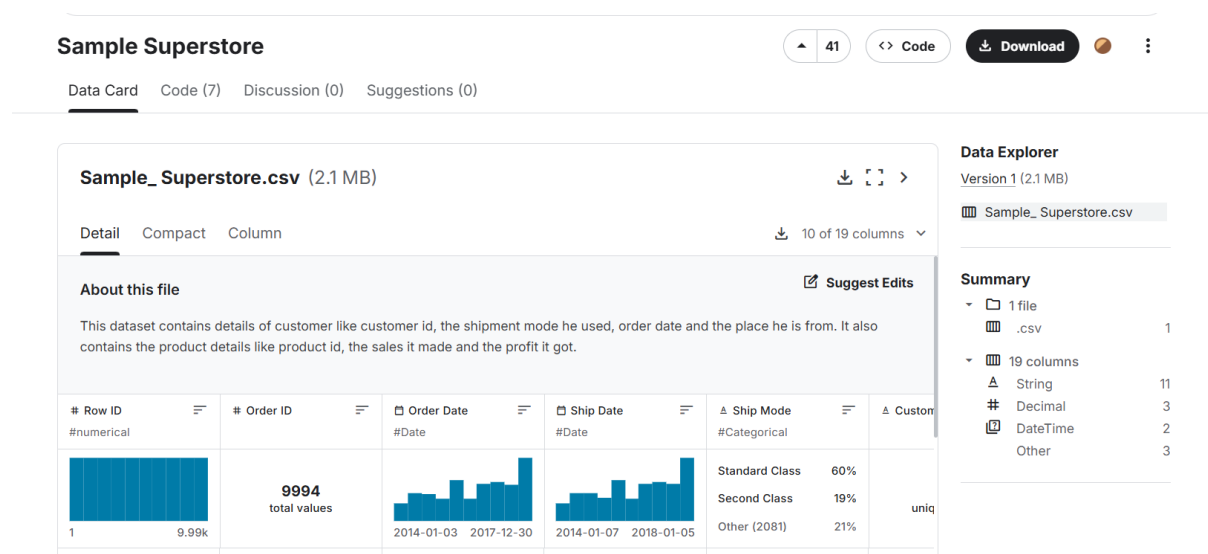


Fig 1 – Dataset source card showing the Sample_ Superstore.csv file (2.1 MB, 19 columns), containing customer, shipment, order, and product/sales details.

Step 2: Data Import

- Opened Power BI Desktop and connected via Get Data → Text/CSV.
- Selected Sample_ Superstore.csv and loaded it via Transform Data, which opened the Power Query Editor.
- Applied Csv.Document with Delimiter = ",", Columns = 19, Encoding = 65001 (UTF-8), and promoted the first row to headers.
- Set correct data types on import: Row ID and Quantity as Whole Number, Sales/Discount/Profit as Decimal, Order Date/Ship Date as Date, and all remaining descriptive fields as Text.

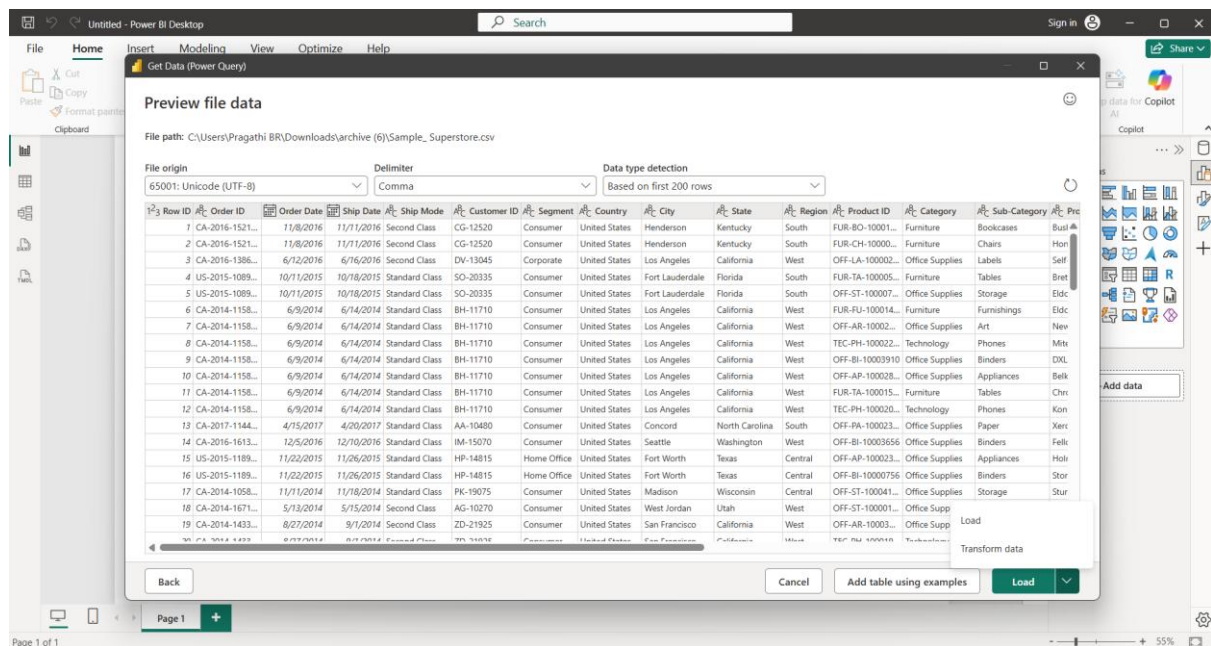


Fig 2 – Power BI Get Data (Power Query) preview window showing the file path, encoding, delimiter settings, and a preview of all 19 source columns before transformation.

Step 3: Data Cleaning and Transformation

- Checked all 19 columns for missing or null values in key fields (Order ID, Product ID, Customer ID) — none found.
- Trimmed leading/trailing spaces (Text.Trim) on 12 text columns: City, Ship Mode, Customer ID, Segment, Country, State, Region, Product ID, Category, Sub-Category, Product Name, Order ID.
- Removed non-printable characters (Text.Clean) from the same 12 text columns to standardize values for reliable grouping and filtering.
- Added derived date columns to support time-intelligence reporting:
 - **Order Year** = Date.Year([Order Date])

- **Month Number** = Date.Month([Order Date])
- **Month Name** = Date.MonthName([Order Date])
- **Year-Month**=Text.From([OrderYear])&"-"&
Text.PadStart(Text.From([Month Number]), 2, "0")

These transformations expanded the working table from 19 to 23 columns and produced a clean, analysis-ready dataset named Sample_ Superstore in the Power BI data model.

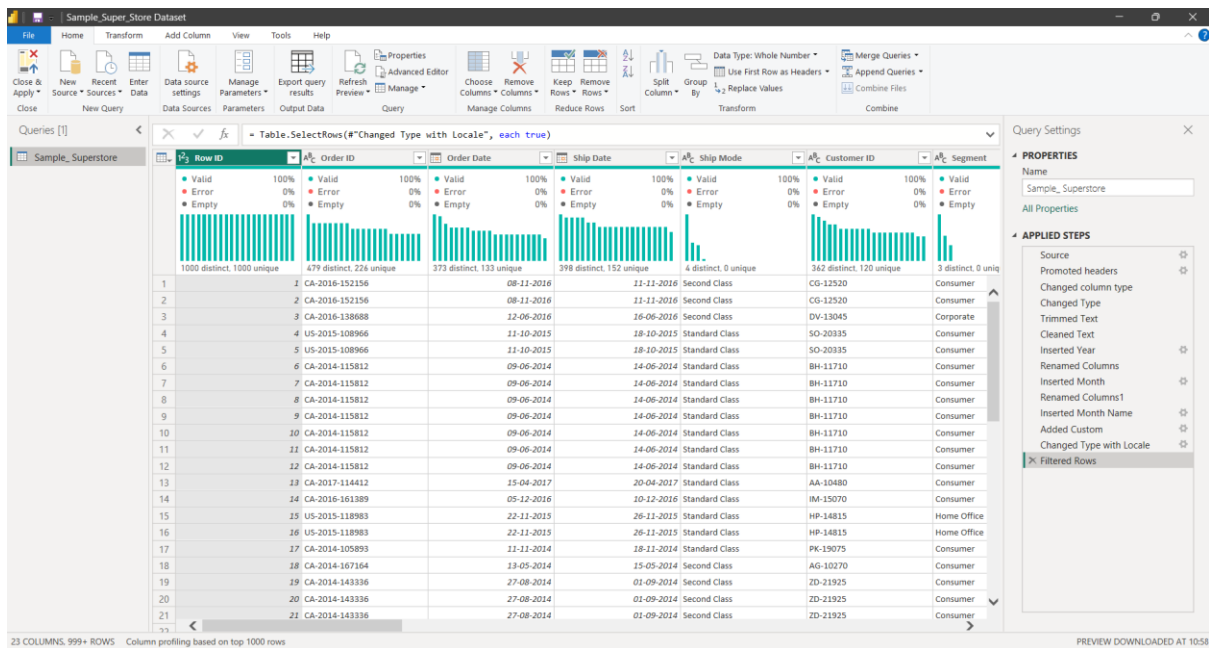


Fig 3 – Power Query Editor showing the full Applied Steps list (right panel) and column-level data quality profiling (Valid/Error/Empty percentages) for the Sample_ Superstore table.

Step 4: Data Modeling

- Loaded the cleaned Sample_ Superstore query into the Power BI data model in Import mode.
- Created a separate **Calendar** date table to support proper time-intelligence analysis, using.
- Calendar =

```

ADDCOLUMNS(
    CALENDAR(
        MIN('Sample_ Superstore'[Order Date]),
        MAX('Sample_ Superstore'[Order Date])
    ),
    "Year", YEAR([Date]),
    "Month Number", MONTH([Date]),
    "Month Name", FORMAT([Date], "MMMM"),
    "Quarter", "Q" & FORMAT([Date], "Q"),

```

)
 "Year-Month", FORMAT([Date], "YYYY-MM")

- Added supporting columns to the Calendar table — Year, Month, Month Name, Quarter, and Day — to enable flexible date-based slicing and DAX time-intelligence functions.

The screenshot shows the Power BI Desktop interface. The 'Table tools' ribbon is active, and the 'Calendar' table is selected. The DAX formula bar shows the following code:

```

1 Calendar =
2 ADDCOLUMNS(
3   CALENDAR(
4     MIN('Sample_Superstore'[Order Date]),
5     MAX('Sample_Superstore'[Order Date])
6   ),
7   "Year", YEAR([Date]),
8   "Month Number", MONTH([Date]),
9   "Month Name", FORMAT([Date], "MMM"),
10  "Quarter", "Q" & FORMAT([Date], "Q"),
11  "Year-Month", FORMAT([Date], "YYYY-MM")
12 )
  
```

The data view below the formula shows a table with the following columns: Date, Year, Month Number, Month Name, Quarter, Year-Month, and Month Year. The data represents the first 18 days of July 2014.

Date	Year	Month Number	Month Name	Quarter	Year-Month	Month Year
01-07-2014 00:00:00	2014	7	July	Q3	2014-07	Jul 2014
02-07-2014 00:00:00	2014	7	July	Q3	2014-07	Jul 2014
03-07-2014 00:00:00	2014	7	July	Q3	2014-07	Jul 2014
04-07-2014 00:00:00	2014	7	July	Q3	2014-07	Jul 2014
05-07-2014 00:00:00	2014	7	July	Q3	2014-07	Jul 2014
06-07-2014 00:00:00	2014	7	July	Q3	2014-07	Jul 2014
07-07-2014 00:00:00	2014	7	July	Q3	2014-07	Jul 2014
08-07-2014 00:00:00	2014	7	July	Q3	2014-07	Jul 2014
09-07-2014 00:00:00	2014	7	July	Q3	2014-07	Jul 2014
10-07-2014 00:00:00	2014	7	July	Q3	2014-07	Jul 2014
11-07-2014 00:00:00	2014	7	July	Q3	2014-07	Jul 2014
12-07-2014 00:00:00	2014	7	July	Q3	2014-07	Jul 2014
13-07-2014 00:00:00	2014	7	July	Q3	2014-07	Jul 2014
14-07-2014 00:00:00	2014	7	July	Q3	2014-07	Jul 2014
15-07-2014 00:00:00	2014	7	July	Q3	2014-07	Jul 2014
16-07-2014 00:00:00	2014	7	July	Q3	2014-07	Jul 2014
17-07-2014 00:00:00	2014	7	July	Q3	2014-07	Jul 2014
18-07-2014 00:00:00	2014	7	July	Q3	2014-07	Jul 2014

Table: Calendar (1,458 rows)

Fig 4 – DAX formula used to create the Calendar table in Power BI, with the resulting Date, Year, Month Number, Month Name, Quarter, and Year-Month columns.

Creating A Modeling:

- Built a **relationship** between the Calendar table and the Sample_ Superstore table, linked via **Date ↔ Order Date**.
- Verified the relationship as **one-to-many**, with the Calendar table positioned as the "one" side (dimension table) and Sample_ Superstore as the "many" side (fact table) — forming a basic star schema.
- Marked the Calendar table as the official **Date Table** in Power BI (Table Tools → Mark as Date Table) so DAX time-intelligence functions (e.g., YTD, MTD, SAMEPERIODLASTYEAR) resolve correctly.
- Validated the model in Model view to confirm the relationship cardinality and cross-filter direction, ensuring accurate aggregation without circular references.

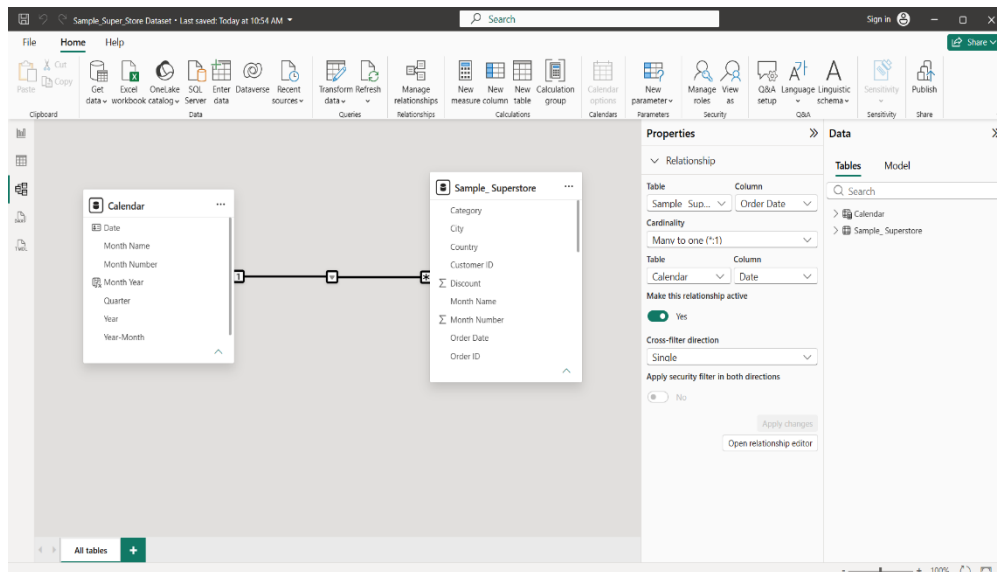


Fig 5 – Power BI Model view showing the one-to-many relationship between the Calendar table and the Sample_Superstore table, linked on Date ↔ Order Date, with cardinality "Many to one" and Single cross-filter direction.

Step 5: DAX Functions Development

Created core DAX measures:

- Total Sales = SUM('Sample_Superstore'[Sales])
- Total Profit = SUM('Sample_Superstore'[Profit])
- Total Orders = DISTINCTCOUNT('Sample_Superstore'[Order ID])
- Profit Margin = DIVIDE([Total Profit], [Total Sales], 0)
- Average Order Value = DIVIDE([Total Sales], [Total Orders], 0)
- Total Quantity = SUM('Sample_Superstore'[Quantity])

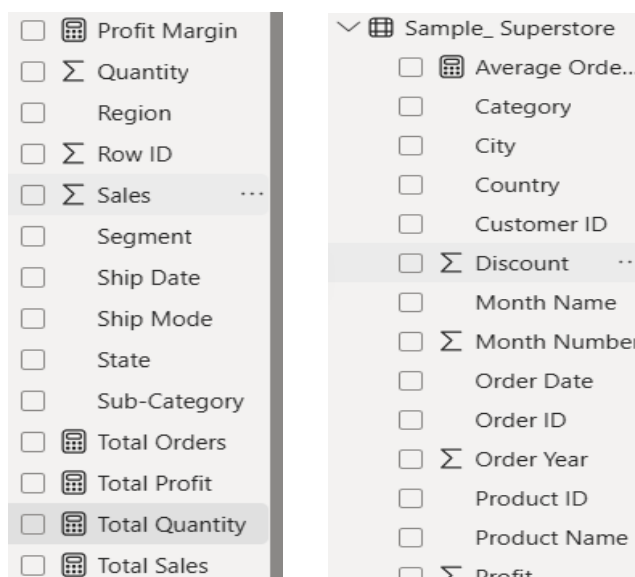


Figure 6: Complete DAX Measures Used in the Superstore Sales Analytics Dashboard

Step 5: Dashboard Development:

Built a three-page interactive dashboard titled "Superstore Sales Analytics" with a top navigation bar linking Executive Overview, Product & Regional Analysis, and Customer & Profitability Analysis.

- Page 1 – Executive Overview:** 4 KPI cards (Total Sales 2.30M, Total Profit 286.40K, Total Orders 5K, Profit Margin 12.5%); a Line chart of Monthly Sales Trend; a donut chart of Sales by Category; slicers for Year, Region, Category, Sub-Category, and Segment (Year slicer now powered by the Calendar table); and a Dataset Information panel (Source: Sample Superstore, Total Records: 9,994, Total Columns: 23) & used APge Navigation Buttons.

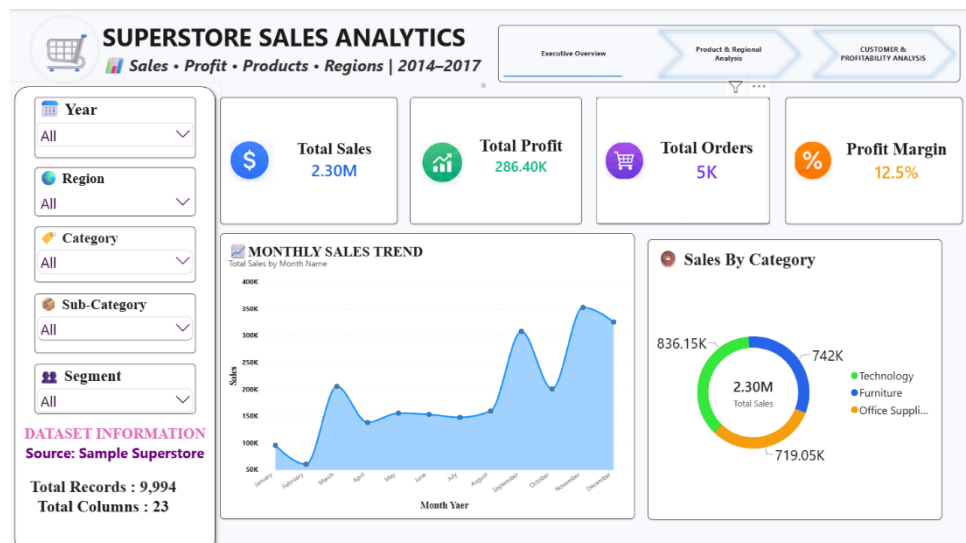


Fig 7: Executive Overview Dashboard – Superstore Sales Analytics

- Page 2 – Product & Regional Analysis:** a horizontal bar chart of Top Sub-Categories by Sales; a treemap of Sales by Region; a map visual of Sales Performance by State; and Region filter buttons for Central, South, East, and West.

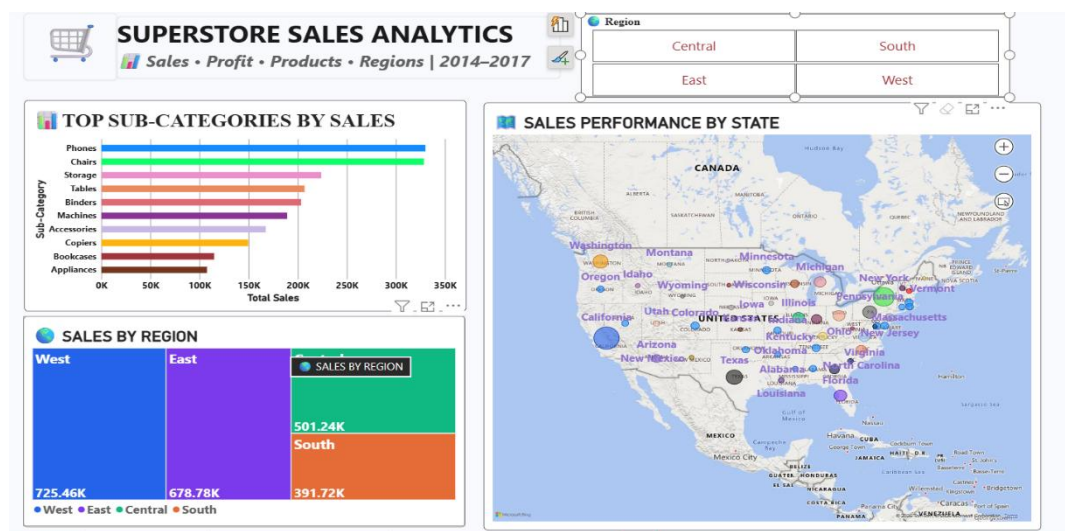


Fig8: Complete Product and Regional Analysis Dashboard

- **Page 3 – Customer & Profitability Analysis:** a pie chart of Sales by Customer Segment; a combo bar chart of Sales & Profit by Category; a Category Performance Summary table; an Average Order KPI card; and a City slicer.

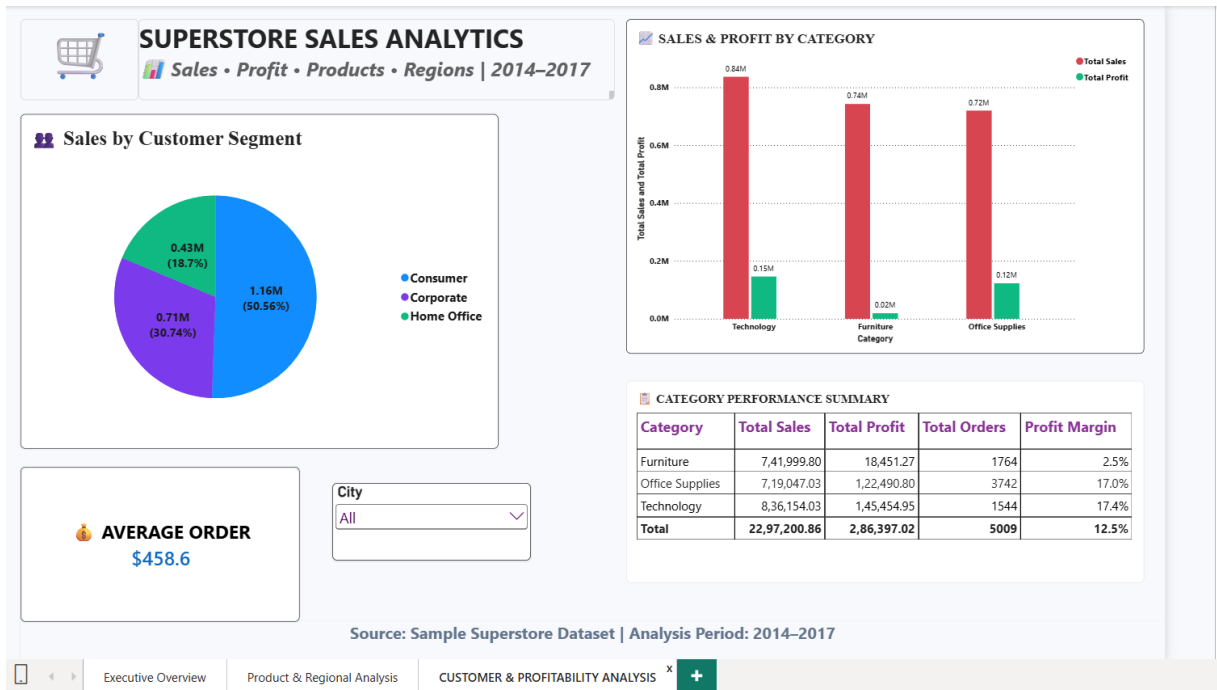


Figure 4: Customer and Profitability Analysis Dashboard

Output

The final Power BI dashboard provides:

- Total Sales (2.30M), Total Profit (286.40K), Total Orders (5K), and Profit Margin (12.5%) at a glance via KPI cards.
- Month-wise and year-wise sales trend, powered by the dedicated Calendar table, showing a strong seasonal peak from September through November.
- Category-wise sales breakdown — Technology leads in sales (836.15K) with the highest profit margin (17.4%), while Furniture lags badly on profitability (2.5% margin despite 742K in sales).
- Region-wise and state-wise sales comparison (West is the top-performing region at 725.46K).
- Sub-category-wise sales ranking (Phones and Chairs are the top two sub-categories).
- Customer segment-wise sales split (Consumer accounts for over half of total sales at 50.56%).
- Average order value (\$458.6) and a full category performance summary table.
- Interactive filtering via Year, Region, Category, Sub-Category, Segment, and City slicers across three report pages.

Category Performance Summary

Category	Total Sales	Total Profit	Total Orders	Profit Margin
Furniture	7,41,999.80	18,451.27	1764	2.5%
Office Supplies	7,19,047.03	1,22,490.80	3742	17.0%
Technology	8,36,154.03	1,45,454.95	1544	17.4%
Total	22,97,200.86	2,86,397.02	5009	12.5%

The dashboard enables management to monitor business performance and make data-driven decisions effectively.

Conclusion

This project successfully imported a single flat-file retail dataset (Sample Superstore CSV) into Power BI, cleaning and standardizing its text fields and enriching it with derived date attributes through Power Query. A dedicated Calendar table was created and linked to the Sample_Superstore table via a one-to-many relationship on Order Date, forming a basic star schema and enabling reliable time-intelligence analysis. DAX measures, KPI cards, charts, slicers, and a three-page navigation structure (Executive Overview, Product & Regional Analysis, Customer & Profitability Analysis) were built to deliver a complete, interactive retail sales and profitability reporting solution.

Learning Outcomes

Through this project, I learned:

- Connecting Power BI to a flat-file (CSV) data source using Get Data → Text/CSV.
- Data cleaning and transformation using Power Query (trim, clean, type correction, derived date columns).
- Creating a dedicated Calendar/Date table using DAX and marking it as the official Date Table for accurate time-intelligence calculations.
- Building and validating a one-to-many relationship between a dimension table (Calendar) and a fact table (Sample_Superstore).
- Writing DAX measures (Total Sales, Total Profit, Total Orders, Profit Margin, Average Order) for KPI cards and visuals.
- Designing a multi-page navigation structure with slicers for interactive filtering.
- Deriving business insights — profitability by category, regional performance, seasonal trends, and customer segmentation — from a modeled retail dataset.